

Q.1. Solve

$$\begin{aligned}x + y + z - 3w &= 1 \\ 2x + 4y + 3z + w &= 3 \\ 3x + 6y + 4z - 2w &= 4.\end{aligned}$$

We can write it as

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 & -3 \\ 2 & 4 & 3 & 1 \\ 3 & 6 & 4 & -2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}}_X = \underbrace{\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}}_B.$$

Consider the augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & -3 & 1 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 6 & 4 & -2 & 4 \end{array} \right].$$

Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & -3 & 1 \\ 0 & 2 & 1 & 7 & 1 \\ 0 & 3 & 1 & 7 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow 2R_3 - 3R_2} \left[\begin{array}{cccc|c} 1 & 1 & 1 & -3 & 1 \\ 0 & 2 & 1 & 7 & 1 \\ 0 & 0 & -1 & -7 & -1 \end{array} \right].$$

Since $\text{Rank } A = \text{Rank } [A | B] = 3$, the system is consistent.

We have

$$\begin{aligned}x + y + z - 3w &= 1 \\ 2y + z + 7w &= 1 \\ -z - 7w &= -1.\end{aligned}$$

$$\Rightarrow z = 1 - 7w, \quad 2y + 1 - 7w + 7w = 1 \Rightarrow y = 0,$$

$$x + 1 - 7w - 3w = 1 \Rightarrow x = 10w.$$

$$\begin{aligned}\text{Solution set} &= \{(10w, 0, 1-7w, w) \mid w \in \mathbb{R}\} \\ &= \{(0, 0, 1, 0) + w(10, 0, -7, 1) \mid w \in \mathbb{R}\}.\end{aligned}$$

Q.2. $S_1 = \{ (1, -1, 2, 0), (3, 0, 0, 1), (2, 1, -1, 0) \}$
 $S_2 = \{ (1, 1, 1, 0), (1, 1, 0, 1), (1, 0, 1, 1), (0, 1, 1, 1), (1, 1, 1, 1) \}.$

Examine whether S_1 and S_2 are L.I. Does S_1 span $\mathbb{R}^4$?

Suppose

$$\alpha_1 (1, -1, 2, 0) + \alpha_2 (3, 0, 0, 1) + \alpha_3 (2, 1, -1, 0) = (0, 0, 0, 0) \text{ for } \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$$

$$\Rightarrow \alpha_1 + 3\alpha_2 + 2\alpha_3 = 0$$

$$-\alpha_1 + \alpha_3 = 0$$

$$2\alpha_1 - \alpha_3 = 0$$

$$\alpha_2 = 0$$

$$\Rightarrow \alpha_1 = 0, \alpha_2 = 0, \alpha_3 = 0.$$

Thus, S_1 is L.I.

2M

Since $\dim \mathbb{R}^4 = 4$ and S_2 has $5(>4)$ elements,

1M

S_2 is L.D.

Again, $\dim \mathbb{R}^4 = 4$ and S_1 has $3(<4)$ elements,

1M

thus S_1 can not span $\mathbb{R}^4$.

Q.3. $V = \{ y \in D(0,1) \mid \frac{dy}{dx} + p(x)y = 0 \},$

& $W = \{ y \in D(0,1) \mid \frac{dy}{dx} + p(x)y = q(x) \},$

Where $p(x)$ and $q(x)$ are non-zero continuous functions on $(0,1)$.
Discuss whether V and W are vector spaces over $\mathbb{R}$ with respect to the usual addition and scalar multiplication.

Since $D(0,1)$ is a vector space over $\mathbb{R}$ with respect to the usual addition and scalar multiplication, it is sufficient to check that whether V and W are subspaces of $D(0,1)$ or not. — 1M.

Let $y_1, y_2 \in V$. Then

$$\begin{aligned} \frac{d}{dx}(y_1 + y_2) + p(x)(y_1 + y_2) &= \frac{dy_1}{dx} + p(x)y_1 + \frac{dy_2}{dx} + p(x)y_2 \\ &= 0 + 0 = 0. \end{aligned}$$

$$\Rightarrow y_1 + y_2 \in V.$$

Let $\alpha \in \mathbb{R}$ and $y \in V$. Then

$$\frac{d}{dx}(\alpha y) + p(x)(\alpha y) = \alpha \left(\frac{dy}{dx} + p(x)y \right) = \alpha \cdot 0 = 0.$$

$\Rightarrow \alpha y \in V$. Thus, V is a vector space over $\mathbb{R}$. 2M.

Since $0 \notin W$, W is not a vector space over $\mathbb{R}$. 1M.

Q4. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $T(x, y, z) = (2x + y + 2z, x + 2y + z)$
 $= (x, y, z) \begin{bmatrix} 2 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix} = XA,$

where $x = (x, y, z)$ & $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

Let $x = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$ Let $X_1 = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}^T$ and $X_2 = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}^T \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$.

Then $T(X_1 + X_2) = (X_1 + X_2)A = X_1A + X_2A = T(X_1) + T(X_2)$

and $T(\alpha X_1) = (\alpha X_1)A = \alpha(X_1A) = \alpha T(X_1)$. ①

Thus, T is linear.

Kernel $T = \{ (x, y, z) \in \mathbb{R}^3 \mid T(x, y, z) = (0, 0) \}$
 $= \{ (x, y, z) \in \mathbb{R}^3 \mid 2x + y + 2z = 0 \text{ and } x + 2y + z = 0 \}$

$\begin{matrix} x + 2y + z = 0 \\ 2x + y + 2z = 0 \end{matrix} \Rightarrow \begin{matrix} x + 2y + z = 0 \\ -3y = 0 \end{matrix} \Rightarrow y = 0 \text{ \& } x = -z$. ①

$\therefore \text{Kernel } T = \{ (-z, 0, z) \mid z \in \mathbb{R} \} = \{ (-1, 0, 1)z \mid z \in \mathbb{R} \}$.

$\therefore$ Basis of Kernel $T = \{ (-1, 0, 1) \}$.

Range $T =$ column space of $\begin{bmatrix} 2 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ OR Row space of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

Basis of range $T = \{ (2, 1), (1, 2) \}$. ①

Nullity $T = \dim \text{Ker } T = 1$ and Rank $T = \dim \text{Range } T = 2$.

$\Rightarrow \text{Rank } T + \text{Nullity } T = 2 + 1 = 3 = \dim \mathbb{R}^3$. ①

Q5

Let $(x, y, z) \in \mathbb{R}^3$ and $(x, y, z) = a(1, 1, 1) + b(0, 1, 1) + c(0, 0, 1)$

$$\Rightarrow a = x \quad a+b = x+y \quad a+b+c = z$$

$$\Rightarrow b = y-x \quad c = z-x-y+x = z-y$$

$$(x, y, z) = x(1, 1, 1) + (y-x)(0, 1, 1) + (z-y)(0, 0, 1)$$

Suppose $\exists$ a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$\text{Then } T(x, y, z) = x \cdot T(1, 1, 1) + (y-x) T(0, 1, 1) + (z-y) T(0, 0, 1)$$

$$= x(0, 1, 1) + (y-x)(0, 0, 1) + (z-y)(1, 1, 1)$$

$$= (z-y, x+z-y, x+y-x+z-y)$$

$$= (z-y, x-y+z, z)$$

Now Yes, $\exists$ a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ satisfying the conditions. (03)

$$T(0, 1, 0) = (-1, -1, 0)$$

$$= -1(0, 1, 0) - 1(1, 0, 0) + 0(0, 0, 1) \quad (1)$$

$$T(1, 0, 0) = (0, 1, 0) = 1(0, 1, 0) + 0(1, 0, 0) + 0(0, 0, 1) \quad (1)$$

$$T(0, 0, 1) = (1, 1, 1)$$

$$= 1(0, 1, 0) + 1(1, 0, 0) + 1(0, 0, 1) \quad (1)$$

$$m(T) = \begin{bmatrix} -1 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Q6

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\det(A) \neq 0$$

$$\Rightarrow A^{-1} \text{ exists.}$$

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ 1 & 0 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)^2(2-\lambda) = 0 \quad \text{--- (*)}$$

$$\Rightarrow 2 - 5\lambda + 4\lambda^2 - \lambda^3 = 0 \quad (1)$$

By ~~Cayley~~ - Cayley - Hamilton Theorem.

$$2I - 5A + 4A^2 - A^3 = 0$$

$$\Rightarrow 2A^{-1} = A^2 - 4A + 5I \quad \text{by multiplying } A^{-1} \text{ both sides.}$$

$$\Rightarrow A^{-1} = \frac{5}{2}I - 2A + \frac{1}{2}A^2$$

$$\text{So } \cancel{R=2} \quad R=-2 \quad (1)$$

By (*), the eigenvalues of A are $1, 1, 2$ (1)

Therefore, the eigenvalues of A^{-1} are $1, 1, \frac{1}{2}$ (1)

Q7 (a) $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 2-\lambda & -1 & -1 \\ -1 & 2-\lambda & -1 \\ -1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)((2-\lambda)^2 - 1) + 1(\lambda - 2 - 1) - 1(1 + 2 - \lambda) = 0$$

$$\Rightarrow (2-\lambda)(3 - 4\lambda + \lambda^2) + \lambda - 3 + \lambda - 3 = 0 \quad (1)$$

$$\Rightarrow 6 - 8\lambda + 2\lambda^2 - 3\lambda + 4\lambda^2 - \lambda^3 + 2\lambda - 6 = 0$$

$$\Rightarrow -\lambda(\lambda^2 - 6\lambda + 9) = 0 \Rightarrow \lambda = 0, 3, 3$$

The eigenvalues of A are $0, 3, 3$

(1)

Eigenvector corresponding to $\lambda=0$

let $x = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \neq 0$ be an eigenvector

Then $Ax = 0$

$$\Rightarrow \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{array}{l} 2a - b - c = 0 \\ -a + 2b - c = 0 \\ -a - b + 2c = 0 \end{array} \right\} a = b = c$$

$\Rightarrow x = R \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $R \neq 0$ is the eigenvector corresponding to $\lambda=0$.

Eigenvector corresponding to $\lambda=3$

let $x = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \neq 0$ be an eigenvector

$$\text{Then } \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 3 \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$\Rightarrow -a - b - c = 0$$

$\Rightarrow x = R \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$, $R \neq 0$ is an eigenvector corresponding to $\lambda=3$

Also $R \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ is an eigenvector corresponding to $\lambda=3$ where $R \neq 0$.

(b)

Algebraic multiplicity of $0 = 1$

Algebraic " of $3 = 2$

Geometric " of $0 = 1$

Geometric " " $3 = 2$

(1)